A. IMPLEMENTATION DETAILS

A.1. Dictionary and fitting constants

The dictionary starts with 72 atoms over eight packet-position windows. We remove 24 direction-related atoms that the diagnostic probes leave invariant. The remaining 48 candidates describe six properties per window: log-length position trend, adjacent log-length variation, covariance between log length and preceding log IAT, and front/back, outer/inner, and alternating-quarter length contrasts. Length and IAT log transforms use training-fitted scales. We normalize each atom by its training median and interquartile range, clip it to $[-10, 10]$, and subtract its training mean. All normalization parameters remain fixed on validation and test inputs.

For induction, $\sigma_t = \max(\mathrm{RMS}(\Delta F_t), 10^{-6})$ in Eq. (2). Each ridge penalty in sparse pursuit and coefficient refitting equals 0.001 times the corresponding mean Gram diagonal. In Eq. (5), $\beta_{\mathrm{nat}} = \beta_{\mathrm{full}} = 0.1$; we floor each normalization scale at 10^{-6} before squaring. We set λ_D to 0.001 times the mean non-intercept data-Hessian diagonal and leave b unpenalized. The class guard uses $\eta = 0.99$ and falls back to zero correction if numerical evaluation reverses the original prediction.

B. A CONDITIONAL GUARANTEE OF RESPONSE-ERROR REDUCTION

CoDiCo reduces response error when the information captured by its learned concepts outweighs decoding bias and class-guard distortion. Theorem B.1 makes this condition explicit for the completed explainer. The analysis uses the formulas returned by Induce and the mixed-loss decoder in Eq. (5). We condition on a finite set of decoder-fitting pairs and fixed training normalization parameters; all scores and concept values are finite.

B.1. Learned concepts expose recoverable response information

Consider P pairs (x_i, x_i') satisfying $c_0(x_i') = c_0(x_i)$. Equation (1) gives $\Delta B_i = 0$, so the classifier response also equals the residual response. Let $q_i = e_{u_i} - e_{v_i}$ encode the classifier's original top-two-class contrast, where e_j is a class-coordinate unit vector. Define $t_i = q_i^\top \Delta F_i$ and $d_i = c(x_i') - c(x_i)$. The contrast stays fixed within each pair. For $p(x) = b + Dc(x)$, its unguarded margin response is $q_i^\top D d_i$; the intercept cancels.

Use the positive weights $\omega_i = 1/(|\mathcal{T}| N_{t(i)} \tau_{t(i)}^2)$ of $\mathcal{L}_{\mathrm{margin}}$. Here $\mathcal{T}$ is the set of probe families, $t(i)$ identifies a pair's family, and N_t is its pair count. The scale τ_t is that family's mean absolute classifier margin response with the numerical floor used in fitting. With column-stacking vectorization $\theta = \mathrm{vec}(D) \in \mathbb{R}^{Ck}$, set $y_i = \sqrt{\omega_i}\, t_i$ and $M_{i:} = \sqrt{\omega_i}\,(d_i^\top \otimes q_i^\top)$. Here $\otimes$ denotes the Kronecker product. The unguarded margin loss is $\|y - M\theta\|_2^2$, and the original explainer's loss is $\mathcal{E}_0 = \|y\|_2^2$.

Lemma B.1 (Captured response energy) *For any returned concept matrix W, let $\Pi = MM^\dagger$, where $\dagger$ denotes the Moore–Penrose pseudoinverse. Define $\theta_0 = M^\dagger y$ and $G = \|\Pi y\|_2^2$. Then every linear decoder satisfies*

$$\|y - M\theta\|_2^2 = \mathcal{E}_0 - G + \|M(\theta - \theta_0)\|_2^2. \tag{B.1}$$

Thus G is the largest reduction of this loss attainable by an unregularized linear decoder on the fixed concepts. Moreover, $G > 0$ if and only if $M^\top y \neq 0$.

Proof. Π is the orthogonal projector onto the column space of M, and $M\theta_0 = \Pi y$. The vectors $(I - \Pi)y$ and $M(\theta - \theta_0)$ are orthogonal. Expanding their squared norms gives Eq. (B.1), with $\|(I - \Pi)y\|_2^2 = \mathcal{E}_0 - G$. The projection vanishes precisely when y is orthogonal to every column of M, equivalently $M^\top y = 0$. □

Response induction seeks classifier-aligned directions within the atom budget. Its fixed-support score in Eq. (4) has a direct regularized fitting interpretation. For the design A, current full-score residual target H, and support $\mathcal{J}$ of Sec. 2.3, fix $w \neq 0$ and $\lambda > 0$. Then

$$\min_v \{\|H - A_{\mathcal{J}} w v^\top\|_F^2 + \lambda \|w\|_2^2 \|v\|_2^2\} = \|H\|_F^2 - Q_{\mathcal{J}}(w).$$

To verify the identity, substitute the minimizer

$$v_* = \frac{H^\top A_{\mathcal{J}} w}{\|A_{\mathcal{J}} w\|_2^2 + \lambda \|w\|_2^2}.$$

Maximizing $Q_{\mathcal{J}}$ therefore maximizes this local regularized reduction. CoDiCo greedily grows supports and refits the selected concepts. Induction uses full-score targets; Lemma B.1 measures the selected-margin information in the resulting formulas. The subsequent guarantee uses their actual G without assuming globally optimal support selection.

B.2. The mixed-loss decoder bounds the cost of response fitting

The actual decoder differs from the pure-response fit θ_0 because it also fits natural scores and full-class responses, with ridge regularization. This difference admits an explicit characterization. For $n \geq 1$ natural training inputs, write $c_\ell = c(x_\ell)$ and $r_\ell = F(x_\ell) - B(x_\ell)$, with means $\bar{c}$ and $\bar{r}$. Assume $\beta_{\mathrm{nat}} > 0$, $\beta_{\mathrm{full}} \geq 0$, and $\lambda_D > 0$, as in the evaluated configuration. Minimizing over the unpenalized intercept gives $b(D) = \bar{r} - D\bar{c}$.

Let $U_{\mathrm{n}}, v_{\mathrm{n}}$ and $U_{\mathrm{f}}, v_{\mathrm{f}}$ be the stacked, scaled designs and targets for the remaining natural and full-response terms:

$$\|U_{\mathrm{n}}\theta - v_{\mathrm{n}}\|_2^2 = \frac{\beta_{\mathrm{nat}}}{nCs_{\mathrm{n}}^2} \sum_{\ell=1}^{n} \|D(c_\ell - \bar{c}) - (r_\ell - \bar{r})\|_2^2, \tag{B.2}$$

$$\|U_{\mathrm{f}}\theta - v_{\mathrm{f}}\|_2^2 = \frac{\beta_{\mathrm{full}}}{PCs_{\mathrm{f}}^2} \sum_{i=1}^{P} \|Dd_i - \Delta r_i\|_2^2. \tag{B.3}$$

Specifically, these designs stack scaled copies of $(c_\ell - \bar{c})^\top \otimes I_C$ and $d_i^\top \otimes I_C$; their target blocks use $r_\ell - \bar{r}$ and Δr_i. The positive scales s_{n} and s_{f} are the natural-residual and full-response RMS normalizers from Eq. (5). All weights, scales, and λ_D retain their fitted values when we eliminate the intercept.

The profiled objective is therefore

$$\begin{aligned} J(\theta) = \|y - M\theta\|_2^2 + \|U_{\mathrm{n}}\theta - v_{\mathrm{n}}\|_2^2 \\ + \|U_{\mathrm{f}}\theta - v_{\mathrm{f}}\|_2^2 + \lambda_D \|\theta\|_2^2. \end{aligned} \tag{B.4}$$

Define the auxiliary curvature $K = U_{\mathrm{n}}^\top U_{\mathrm{n}} + U_{\mathrm{f}}^\top U_{\mathrm{f}} + \lambda_D I$ and target vector $h = U_{\mathrm{n}}^\top v_{\mathrm{n}} + U_{\mathrm{f}}^\top v_{\mathrm{f}}$, and set $\mathcal{H} = M^\top M + K$.

Lemma B.2 (Mixed-loss decoder bias) *Equation* (B.4) *has the unique minimizer*

$$\theta_* = \mathcal{H}^{-1}(M^\top y + h). \tag{B.5}$$

With $\zeta = h - K\theta_0$, its excess margin loss over the pure-response optimum is

$$\varepsilon_{\mathrm{dec}} = \|M\mathcal{H}^{-1}\zeta\|_2^2 \leq \zeta^\top \mathcal{H}^{-1} \zeta. \tag{B.6}$$

For numerically returned coefficients $\widehat{\theta}$, define $\rho = \mathcal{H}\widehat{\theta} - (M^\top y + h)$ and $\varepsilon_{\mathrm{opt}} = \rho^\top \mathcal{H}^{-1} \rho$. They satisfy

$$\|y - M\widehat{\theta}\|_2 \leq \sqrt{\mathcal{E}_0 - G + \varepsilon_{\mathrm{dec}}} + \sqrt{\varepsilon_{\mathrm{opt}}}. \tag{B.7}$$

Proof. Since $K \succeq \lambda_D I \succ 0$, the normal equations have the unique solution in Eq. (B.5). The least-squares equations give $M^\top M\theta_0 = M^\top y$. Subtracting them from the decoder equations yields $\theta_* - \theta_0 = \mathcal{H}^{-1}\zeta$. Lemma B.1 identifies its excess margin loss as $\|M\mathcal{H}^{-1}\zeta\|_2^2$. Because $M^\top M \preceq \mathcal{H}$, this quantity is at most $\zeta^\top\mathcal{H}^{-1}\zeta$. Also, $\widehat{\theta} - \theta_* = \mathcal{H}^{-1}\rho$, so $\|M(\widehat{\theta} - \theta_*)\|_2 \le \sqrt{\varepsilon_{\mathrm{opt}}}$. The triangle inequality and Eq. (B.1) give Eq. (B.7). □

The vector ζ measures how auxiliary supervision and regularization displace the minimum-norm response fit. If $h = K\theta_0$, the mixed objective retains that fit and $\varepsilon_{\mathrm{dec}} = 0$. Equation (B.6) also provides a conservative upper bound when this compatibility is approximate. All subsequent sufficient conditions remain valid when $\zeta^\top\mathcal{H}^{-1}\zeta$ replaces $\varepsilon_{\mathrm{dec}}$. The analysis therefore retains Recover's auxiliary supervision and ridge penalty.

B.3. Captured energy can dominate decoding and guard costs

The class guard changes the response at both paired inputs. Let $p(x) = \widehat{b} + \widehat{D}c(x)$ be the returned correction, with $\widehat{\theta} = \mathrm{vec}(\widehat{D})$, and let $a_i = a(x_i)$ and $a'_i = a(x'_i)$. Define the margin response removed by guarding as

$$\xi_i = q_i^\top\big[(1 - a'_i)p(x'_i) - (1 - a_i)p(x_i)\big], \tag{B.8}$$

and its weighted magnitude as $\gamma = (\sum_i \omega_i\xi_i^2)^{1/2}$. This term includes any intercept contribution created by different guard multipliers. For the completed explainer $S = B + ap$, write $\mathcal{E}_S = \sum_i \omega_i\big(t_i - q_i^\top\Delta S_i\big)^2$.

Theorem B.1 (Response-error reduction) *Under the preceding assumptions, CoDiCo's guarded output satisfies*

$$\sqrt{\mathcal{E}_S} \le \sqrt{\mathcal{E}_0 - G + \varepsilon_{\mathrm{dec}}} + \delta, \qquad \delta = \sqrt{\varepsilon_{\mathrm{opt}}} + \gamma. \tag{B.9}$$

In particular, if $\mathcal{E}_0 > 0$, $\delta < \sqrt{\mathcal{E}_0}$, and

$$G > \varepsilon_{\mathrm{dec}} + 2\delta\sqrt{\mathcal{E}_0} - \delta^2, \tag{B.10}$$

then $\mathcal{E}_S < \mathcal{E}_0$. With an exact decoder solve and inactive guards, $G > \varepsilon_{\mathrm{dec}}$ is sufficient for strict improvement.

Proof. Concept preservation gives $\Delta B_i = 0$. Therefore $q_i^\top\Delta S_i = q_i^\top\widehat{D}d_i - \xi_i$. The weighted final error vector equals $y - M\widehat{\theta} + (\sqrt{\omega_i}\xi_i)_i$. Its norm is at most $\|y - M\widehat{\theta}\|_2 + \gamma$. Lemma B.2 gives Eq. (B.9). Under Eq. (B.10), $\mathcal{E}_0 - G + \varepsilon_{\mathrm{dec}} < (\sqrt{\mathcal{E}_0} - \delta)^2$. Taking square roots and adding δ proves strict improvement. □

The guard term has an per-input upper bound:

$$\gamma \le \left[\sum_i \omega_i\left((1 - a'_i)|q_i^\top p(x'_i)| + (1 - a_i)|q_i^\top p(x_i)|\right)^2\right]^{1/2}. \tag{B.11}$$

For a unit requested correction, the guard permits $a(x) = 1$ in exact arithmetic whenever

$$p_j(x) - p_y(x) \le \eta\big(B_y(x) - B_j(x)\big) \quad \text{for all } j, \tag{B.12}$$

where y is the original explainer's winning class. If this condition holds at every input, then $\gamma = 0$. Otherwise, Eq. (B.11) accounts for the response cost of preserving the original decisions. Equation (B.9) continues to use the returned multipliers when the numerical fallback sets a correction to zero.

B.4. A separate sufficient condition controls the reported ratio

The reported ratio R uses pooled absolute error, whereas $\mathcal{E}_S$ uses the decoder's family-normalized squared error. Let $\mu = P^{-1}\sum_i |t_i| > 0$ and $\kappa_\omega = (P^{-2}\sum_i \omega_i^{-1})^{1/2}$. Here μ is the original explainer's margin-response MAE on these pairs.

Corollary B.1 (Absolute-error ratio) *On the same pairs as Theorem B.1,*

$$R \le \frac{\kappa_\omega}{\mu}\left(\sqrt{\mathcal{E}_0 - G + \varepsilon_{\mathrm{dec}}} + \delta\right). \tag{B.13}$$

Consequently, $R < 1$ whenever the parenthesized term is less than μ/κ_ω.

Proof. For $e_i = t_i - q_i^\top\Delta S_i$, weighted Cauchy–Schwarz gives

$$\frac{1}{P}\sum_i |e_i| = \sum_i \frac{\sqrt{\omega_i}|e_i|}{P\sqrt{\omega_i}} \le \kappa_\omega\sqrt{\mathcal{E}_S}. \tag{B.14}$$

Divide by μ and apply Eq. (B.9). □

The guarantee identifies a concrete route to improvement: the returned concepts capture classifier-response information, the response-aligned decoder retains enough of that information, and the class guard leaves enough of the correction intact. For example, $G = \mathcal{E}_0$, $\zeta = 0$, an exact solve, and inactive guards imply zero response error. Partial capture also guarantees improvement when Eq. (B.10) holds. Corollary B.1 supplies the stronger condition needed for the absolute-error ratio.

These are deterministic guarantees on the decoder-fitting pairs, with no independence or optimal-support assumption. The projection identity applies to any fixed pair set, but the normal-equation bound above uses the fitting objective on those same pairs. Extending the guarantee to unseen traffic requires additional control of generalization or distribution shift. The held-out results in Sec. 3 measure empirical recovery; we have not certified those results by evaluating the sufficient conditions above.